

Dharanesh Palaniappan

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Research Profile

Final-year M.Sc. Physics student conducting observational and data-driven research in solar and heliospheric physics. Experienced in analyzing SDO/AIA and SDO/HMI observations. Research interests include solar physics, heliophysics, and astronomical data analysis.

Manuscripts % Preprints

Palaniappan, D. & Katuwal, K. (2026). *Multi-Instrument Analysis of NOAA AR 12781: Coupling Surface Evolution of the AR with Its In-Situ Solar Wind Signatures*. arXiv preprint, arXiv:2609.13689. First-author manuscript under review.

Palaniappan, D. (2025). *Preliminary Analysis of a Potential Asteroid from the IASC Campaign Using Pan-STARRS Data*. arXiv preprint, arXiv:2506.02963.

Research Projects

Multi-Observable Characterization of Solar ARs, CHs, and the QS

In Preparation

- Developing and testing a Python/SunPy pipeline to analyze SDO/AIA, SDO/HMI observations.
- Comparing coronal emission, magnetic-field and flux distributions, and velocity distributions across active regions, coronal holes, and quiet-Sun regions.

Morphological Analysis of a Galaxy

IUCAA Workshop

- Analyzed HST FITS images using SAOImage DS9 and constructed a growth curve from background-subtracted cumulative flux.
- Estimated an angular radius of approximately 2.4 arcsec and a physical radius of approximately 15.3 kpc.
- Calculated photometric properties: Apparent Magnitude ≈ 21.06 and Absolute Magnitude ≈ -21.39 .

Analysis of 501 Stars and Their Exoplanetary Systems

Independent work

- Analyzed Kepler and TESS light curves for 501 stars selected from the MAST database using Lightcurve.
- Searched for transit-like signals and compared candidates with confirmed exoplanet systems using the NASA Exoplanet Archive.

Research Internship: Modern Materials and Devices

IMMD, Chennai

- Gained hands-on experience in nanomaterial synthesis and crystal growth methods.
- Analyzed spectral datasets (XRD, FT-IR, UV-DRS, NIR) and evaluated electrochemical performance (CV, GCD).
- Used OriginPro for scientific visualization, curve fitting, and multi-peak analysis.

Education

Madras Christian College (University of Madras) | Chennai, India

June 2025 – May 2027 (expected)

Master of Science (M.Sc.) in Physics | CGPA: 7.55/10 (till sem 2)

Madras Christian College (University of Madras) | Chennai, India

July 2022 – April 2025

Bachelor of Science (B.Sc.) in Physics | CGPA: 8.06/10

Technical Skills

Programming and analysis	Python; SunPy; PySPEDAS; Lightkurve; NumPy; Astropy; pandas; Matplotlib
Solar and astronomy data	SDO/AIA; SDO/HMI (line-of-sight and vector magnetograms, Dopplergrams, continuum images); OMNI solar-wind data; MAST
Imaging and observation	SAOImage DS9; Astrometrica; Celestron software; Telescope handling

Summer Schools & Workshops

The South Asian Astrophysics Summer School

July 2025

- Analyzed real-time solar data using both remote and in situ measurement techniques.
- Worked with data from the Parker Solar Probe, HMI, and AIA instruments.

Intro2Astro

September 2025

- Received introductory training in Python and astronomical data analysis using real observational data.
- Learned about research and career opportunities in astronomy.

Sagan Summer Workshop

July 2025

- Received training in exoplanet detection and characterization, including light-curve analysis, transit methods, and statistical tools.

Leadership & Observational Experience

Citizen Scientist | International Astronomical Search Collaboration

February 2025 – March 2026

- Analyzed astronomical images via Astrometrica to detect asteroids for NASA-affiliated citizen science initiatives.
- Refined practical skills in astrometry, data analysis, and data reduction.

Secretary | B.M. Arthur Astro-Ephemeris Club (MCC)

July 2023 – April 2025

- Organized and led campus astronomy activities, stargazing workshops, and celestial-event viewings.
- Managed setup, alignment, and operation of Celestron systems and associated control software.

Observatory Volunteer & Assistant | Devadas Observatory

2022 – 2025

- Assisted observatory staff with telescope handling, routine maintenance, and practical night-sky navigation.
- Operated telescopes, trained peers, and guided more than 100 visitors during public observing sessions.

Honors

International Astronomy and Astrophysics Competition | Silver Honor

2023

Selected Courses

Confronting The Big Questions: Highlights of Modern Astronomy

University of Rochester

Astronomy: Exploring Time and Space

University of Arizona

The Science of The Solar System

Caltech

The Diversity of Exoplanets

University of Geneva